

3x3 Array Multiplier

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Abstract— This project presents a 3×3 binary multiplier that multiplies two 3-bit numbers to produce a 6-bit output. The circuit is built using basic logic gates and partial product addition. Transient analysis was performed to observe how the outputs change over time and to verify correct timing and logic levels. The simulation results confirm that the circuit works correctly and can be used in simple digital systems for multiplication.

Keywords— 3×3 Array Multiplier, Binary Multiplication, Logic Gates, Digital Circuit, Partial Product

I. INTRODUCTION

Multiplication is one of the basic operations in digital systems, and it is widely used in processors, calculators, and arithmetic circuits. A 3×3 array multiplier multiplies two 3-bit binary numbers to produce a 6-bit output using basic logic gates like AND, OR, and XOR. The circuit works by generating partial products and then adding them in stages to get the final result. This project focuses on designing and simulating the multiplier to verify correct outputs and timing, making it a useful exercise in understanding digital arithmetic operations.

II. OBJECTIVES

The main objective of this project is to design and simulate a 3×3 binary array multiplier using basic logic gates. The goal is to verify that the circuit correctly produces a 6-bit output for different input combinations. Transient analysis is carried out to observe the timing behavior and confirm the proper working of the circuit. This helps in understanding how digital multipliers are built and applied in simple arithmetic systems.

III. IMPLEMENTATION

The 3×3 binary array multiplier was implemented using basic logic gates to generate partial products and adders to combine them into the final result. Two sets of 3-bit binary inputs were applied, and AND gates were used to produce the partial products. These partial products were then added using half adders and full adders arranged in an array structure. The circuit was designed and simulated in eSim, and transient analysis was performed with pulse inputs to observe the output changes over time. The simulation results confirmed that the multiplier correctly produced a 6-bit output for all input combinations.

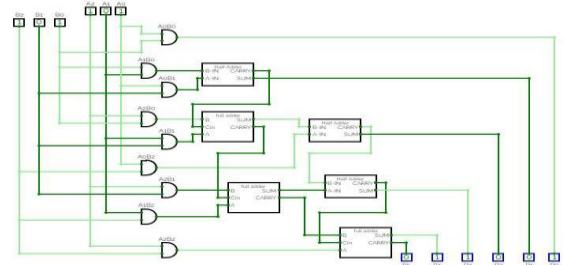


Fig. 1. Logic Diagram

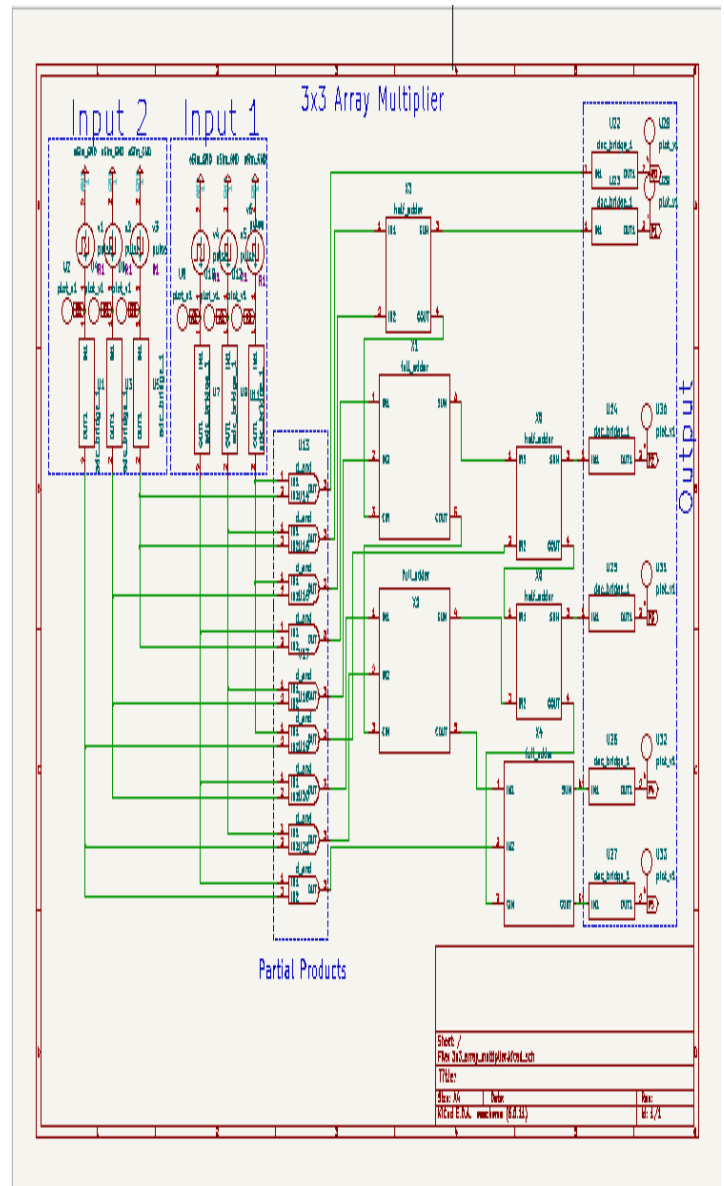


Fig. 2. eSim Circuit Schematic

IV. RESULTS

The simulation of the 3×3 binary array multiplier was successfully carried out in eSim. The transient analysis showed that the outputs changed correctly with respect to the applied input pulses. For different combinations of the two 3-bit inputs (A0, A1, A2 and B0, B1, B2), the circuit produced the expected 6-bit output (P0, P1, P2, P3, P4, P5). This confirms that the multiplier design works correctly and performs binary multiplication as intended.

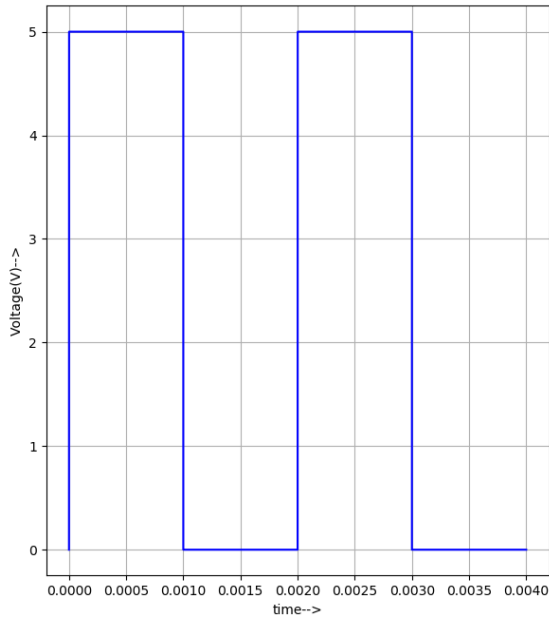


Fig. 3. Input A1

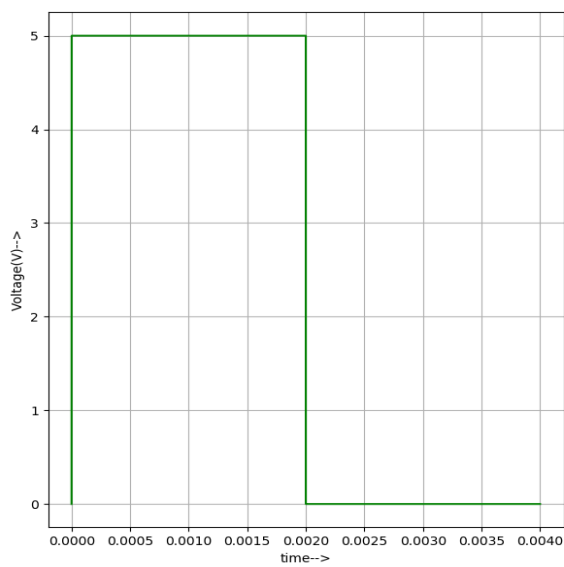


Fig. 4. Input A2

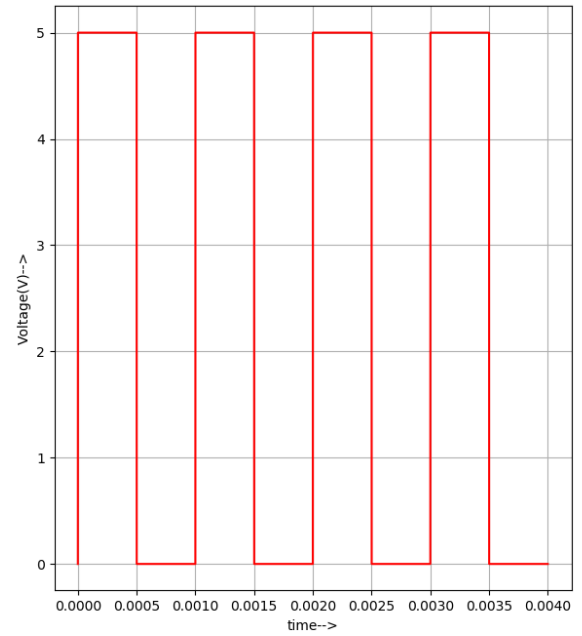


Fig. 5. Input A0

Figures 3, 4, and 5 represent the three input bits of A, where Fig. 3 shows A1, Fig. 4 shows A2, and Fig. 5 shows A0.

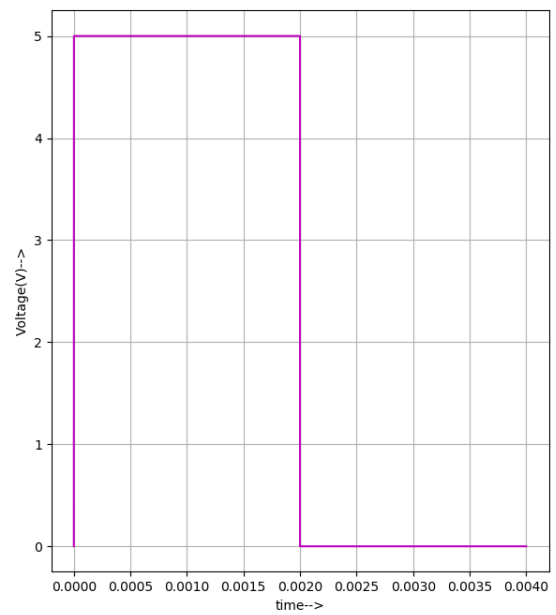


Fig. 6. Input B2

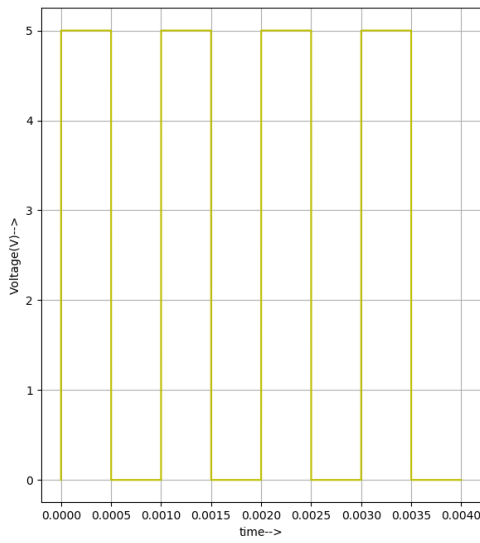


Fig. 7. Input B0

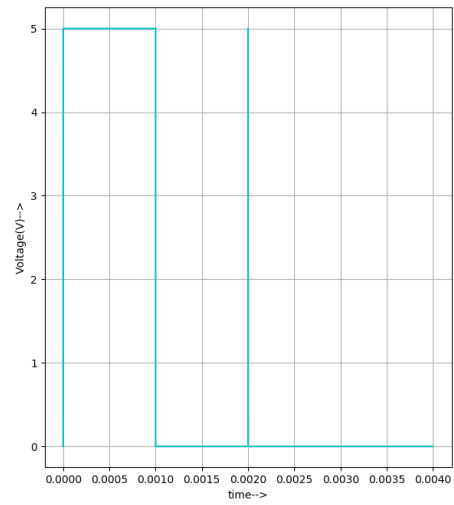


Fig. 9. Output P5

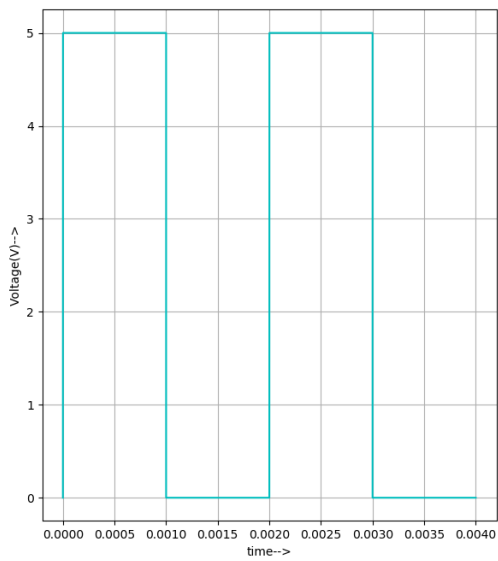


Fig. 8. Input B1

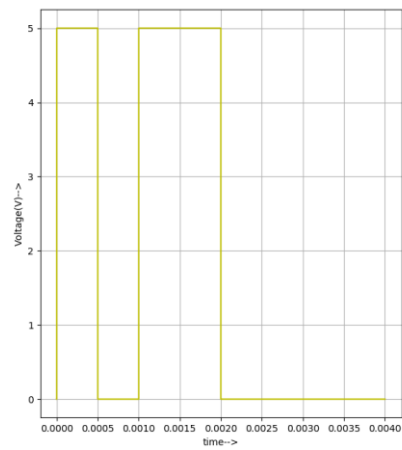


Fig. 10. Ouput P4

Figures 6, 7, and 8 represent the three input bits of B, where Fig. 6 shows B2, Fig. 7 shows B0, and Fig. 8 shows B1.

Fig. 13. Ouput P1

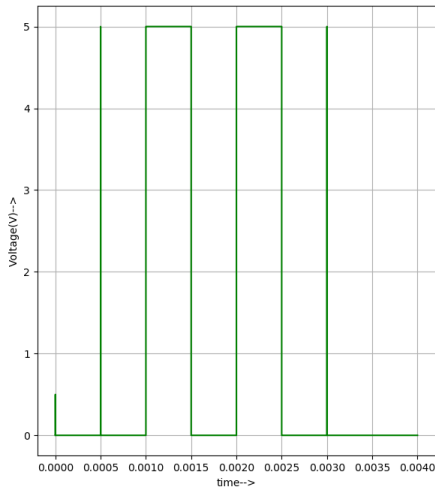


Fig. 11. Ouput P3

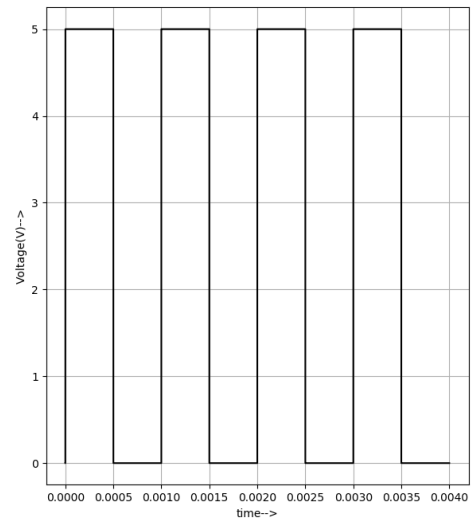


Fig. 14. Ouput P0

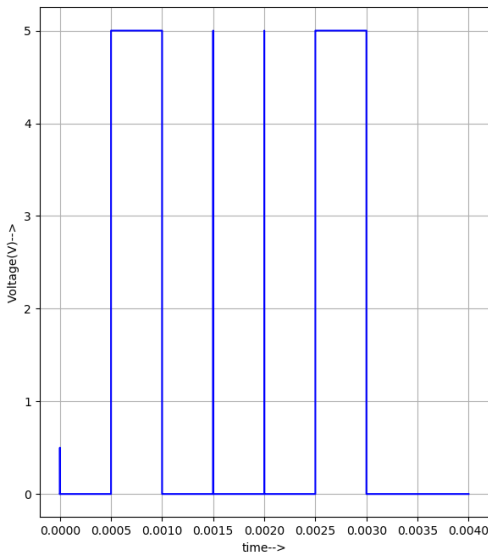
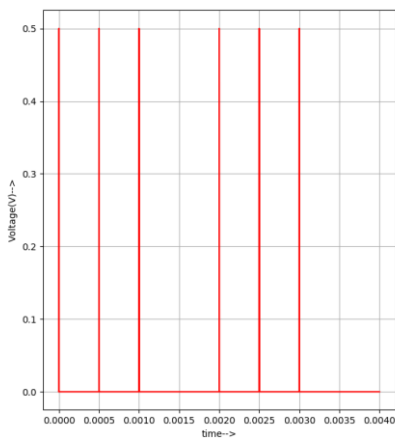


Fig. 12. Ouput P2



Figures 9 to 14 represent the six output bits of the multiplier, where each figure corresponds to one output from P0 to P5.

V. ANALYSIS

The working of the 3×3 binary array multiplier was verified using transient analysis. This analysis showed how the outputs (P0–P5) changed over time when different input pulses (A0–A2 and B0–B2) were applied. The waveforms confirmed proper timing, rise and fall of signals, and the correct generation of the 6-bit product for all tested inputs.

VI. CONCLUSION

The 3×3 binary array multiplier was successfully migrated in eSim using basic logic gates. Transient analysis confirmed that the circuit produced the correct 6-bit output (P0–P5) for different input combinations. The results showed proper timing behavior and accurate multiplication. This design demonstrates a simple and effective method of implementing multiplication in digital systems.

REFERENCES

D. T. Raju and M. D. S. Manasa, "Design and Implementation of 3*3 Array Multiplier using DPTL Logic," *International Journal of Engineering Research & Technolog*